

Supplemental notes for Lecture #110

Cardiogenetics/Pulmonary Genetics

Slide – 16: Aortic Root Dilatation/Dissection

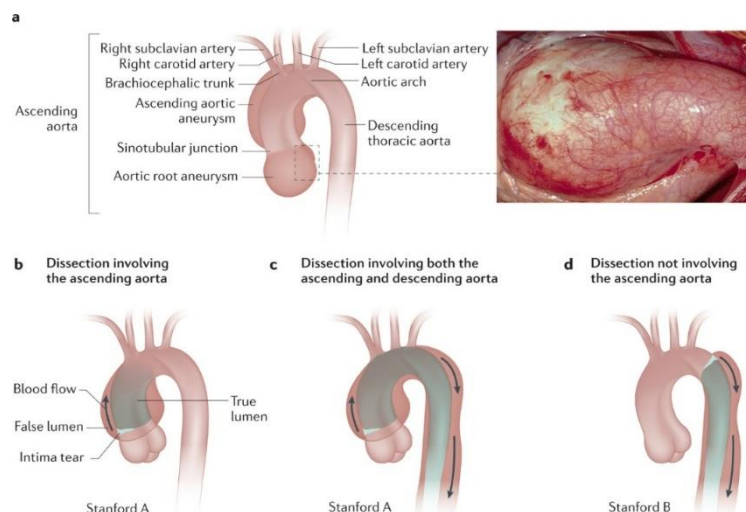
Aortic Root Dilatation/Dissection or a **Family History** of Marfan Syndrome plus a Combination of **Systemic Findings (>7 Points)**

- | | |
|---------------------------------|---|
| ○ Wrist & Thumb Sign | 3 |
| ○ Pectus Carinatum | 2 |
| ○ Hindfoot deformity | 2 |
| ○ Pneumothorax | 2 |
| ○ Protrusio acetabulae | 2 |
| ○ Scoliosis/TL Kyphosis | 1 |
| ○ 3 of 5 Facial Features | 1 |
| ○ Myopia | 1 |
| ○ Wrist or Thumb Sign | 1 |
| ○ Pectus Excavatum/Asymmetry | 1 |
| ○ Pes Planus | 1 |
| ○ Dural Ectasia | 2 |
| ○ ↓Upper:Lower Seg & ↑AS:Height | 1 |
| ○ Reduced Elbow Extension | 1 |
| ○ Skin Striae | 1 |
| ○ Mitral Valve Prolapse | 1 |

(Without atypical features suggesting alternative diagnoses)

Slide – 16: MARFAN AORTIC DISSECTION CLASSIFICATION

- Source: *Nat Rev Dis Primers*. 2021 Sep 2; 7(1): 64



Slide – 20: EDS CLASSIFICATION

- Source: American Journal of Medical Genetics Part C (Seminars in Medical Genetics) 175C:8–26 (2017)

TABLE I. Clinical Classification of the Ehlers-Danlos Syndromes, Inheritance Pattern, and Genetic Basis

	Clinical EDS subtype	Abbreviation	IP	Genetic basis	Protein
1	Classical EDS	cEDS	AD	Major: <i>COL5A1</i> , <i>COL5A1</i> Rare: <i>COL1A1</i> c.934C>T, p.(Arg312Cys)	Type V collagen Type I collagen
2	Classical-like EDS	clEDS	AR	<i>TNXB</i>	Tenascin XB
3	Cardiac-valvular	cvEDS	AR	<i>COL1A2</i> (biallelic mutations that lead to <i>COL1A2</i> NMD and absence of pro $\alpha 2(I)$ collagen chains)	Type I collagen
4	Vascular EDS	vEDS	AD	Major: <i>COL3A1</i> Rare: <i>COL1A1</i> c.934C>T, p.(Arg312Cys) c.1720C>T, p.(Arg574Cys) c.3227C>T, p.(Arg1093Cys)	Type III collagen Type I collagen
5	Hypermobile EDS	hEDS	AD	Unknown	Unknown
6	Arthrochalasia EDS	aEDS	AD	<i>COL1A1</i> , <i>COL1A2</i>	Type I collagen
7	Dermatosparaxis EDS	dEDS	AR	<i>ADAMTS2</i>	ADAMTS-2
8	Kyphoscoliotic EDS	kEDS	AR	<i>PLOD1</i> <i>FKBP14</i>	LH1 FKBP22
9	Brittle Cornea syndrome	BCS	AR	<i>ZNF469</i> <i>PRDM5</i>	ZNF469 PRDM5
10	Spondylodysplastic EDS	spEDS	AR	<i>B4GALT7</i> <i>B3GALT6</i> <i>SLC39A13</i>	$\beta 4$ GalT7 $\beta 3$ GalT6 ZIP13
11	Musculocontractural EDS	mcEDS	AR	<i>CHST14</i> <i>DSE</i>	D4ST1 DSE
12	Myopathic EDS	mEDS	AD or AR	<i>COL12A1</i>	Type XII collagen
13	Periodontal EDS	pEDS	AD	<i>C1R</i> <i>C1S</i>	C1r C1s

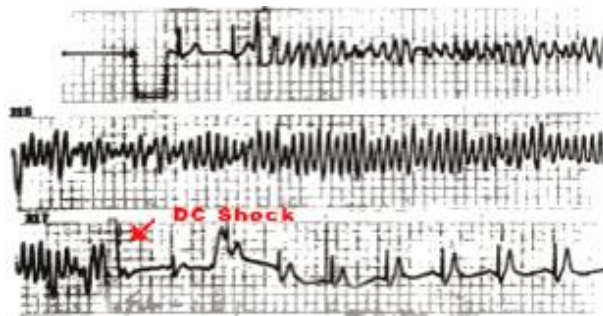
IP, inheritance pattern; AD, autosomal dominant; AR, autosomal recessive, NMD, nonsense-mediated mRNA decay.

Slide – 21: EDS - VALIDATED HYPERMOBILITY SCALES**THE BEIGHTON SCALE (BS)**

- 9-point scale → Beighton Score
 - Passive dorsiflexion of the little fingers beyond 90° (one point each finger)
 - Passive apposition of the thumbs to the flexor aspect of the forearm (one point each thumb)
 - Hyperextension of the elbows 10° beyond 180° (one point each elbow)
 - Hyperextension of the knees 10° beyond 180° (one point each knee)
 - Forward flexion of the trunk with knees fully extended so that the palms of the hand rest flat on the floor (one point)

Slide – 27: LQTS

- **EKG in LQTS before and after Defibrillation**



- **Drugs to avoid in LQTS:** <https://www.sads.org.uk/drugs-to-avoid/>
- **Long QT Syndrome: Genetic Causes** (<https://www.ncbi.nlm.nih.gov/books/NBK1129/>):

Gene ¹	% of LQTS Attributed to Pathogenic Variants in Gene ²	Gene Validity Classification ³	LQTS Phenotype(s)	Comment	OMIM Phenotype
KCNQ1	30%-35%	Definitive	LQTS w/broad-based T waves ⁴ w/ or w/o atrial fibrillation	Note: Biallelic variants in KCNQ1 cause Jervell and Lange-Nielsen	192500

Gene ¹	% of LQTS Attributed to Pathogenic Variants in Gene ²	Gene Validity Classification ³	LQTS Phenotype(s)	Comment	OMIM Phenotype
				syndrome	
<i>KCNH2</i>	25%-30%	Definitive	LQTS w/bifid T waves ⁴		613688
<i>SCN5A</i> (GoF variants)	5%-10%	Definitive	LQTS w/long horizontal ST segment ⁴		603830
<i>CACNA1C</i> (GoF variants)	<1%	Definitive	LQTS	Some persons have additional features consistent w/ Timothy syndrome .	601005
<i>CALM1</i>	<1%	Definitive	LQTS or LQTS-CPVT overlap ⁵		616247
<i>CALM2</i>	<1%	Definitive			616249
<i>CALM3</i>	<1%	Definitive			618782
<i>CAV3</i>	<1%	Definitive	LQTS	Caveolinopathy is assoc w/a broad spectrum of phenotypes. ⁶	123320 192600 606072 611818 614321
<i>KCNJ2</i>	<1%	Definitive	LQTS	Some persons have additional features consistent w/ Andersen-Tawil syndrome .	170390
<i>TRDN</i>	<1%	Definitive ⁷	CPVT-LQTS overlap syndrome ⁸		615441
<i>KCNE1</i> ⁹	<1%	Strong	Susceptibility to acquired LQTS ⁶	Note: Biallelic variants in <i>KCNE1</i> cause Jervell and Lange-Nielsen syndrome	613695
		Limited	LQTS		
<i>KCNE2</i>	<1%	Strong	Susceptibility to acquired LQTS ³		613693

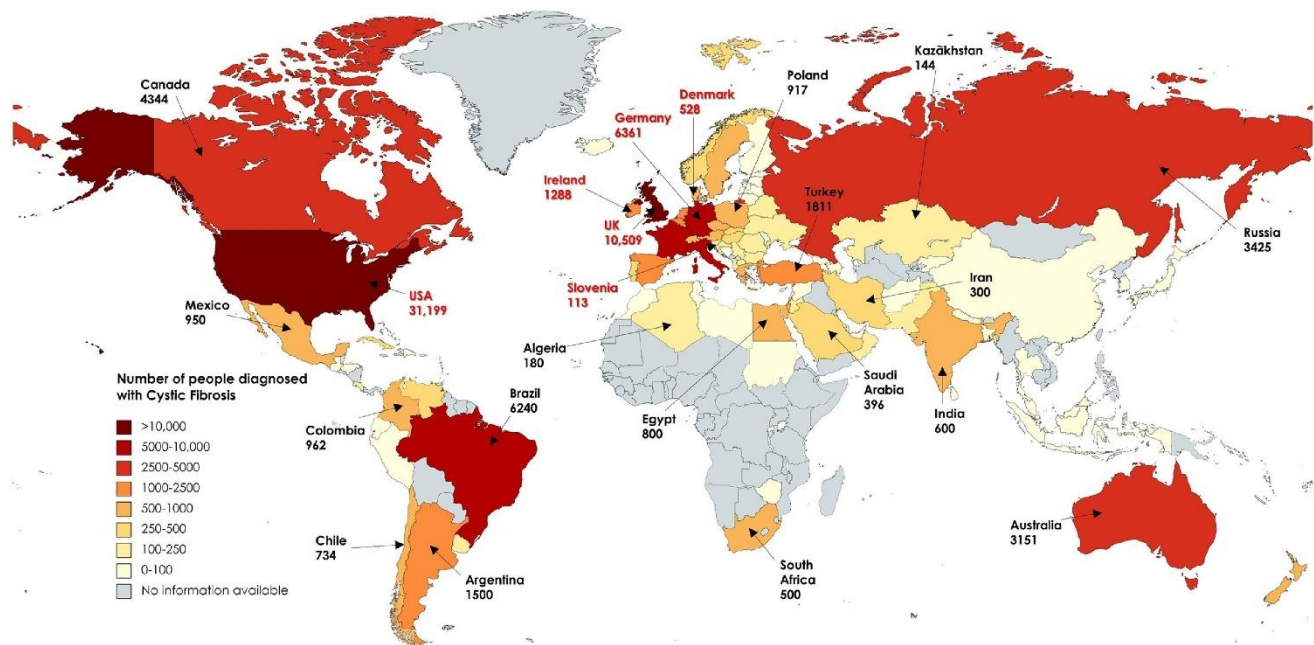
Slide – 28: Brugada Syndrome:

- **Coding genes**
 - Sodium current (I_{Na}): *FGF12*, *GPD1L*, *PKP2*, *RANGRF*, *SCN1B*, *SCN2B*, *SCN3B*,

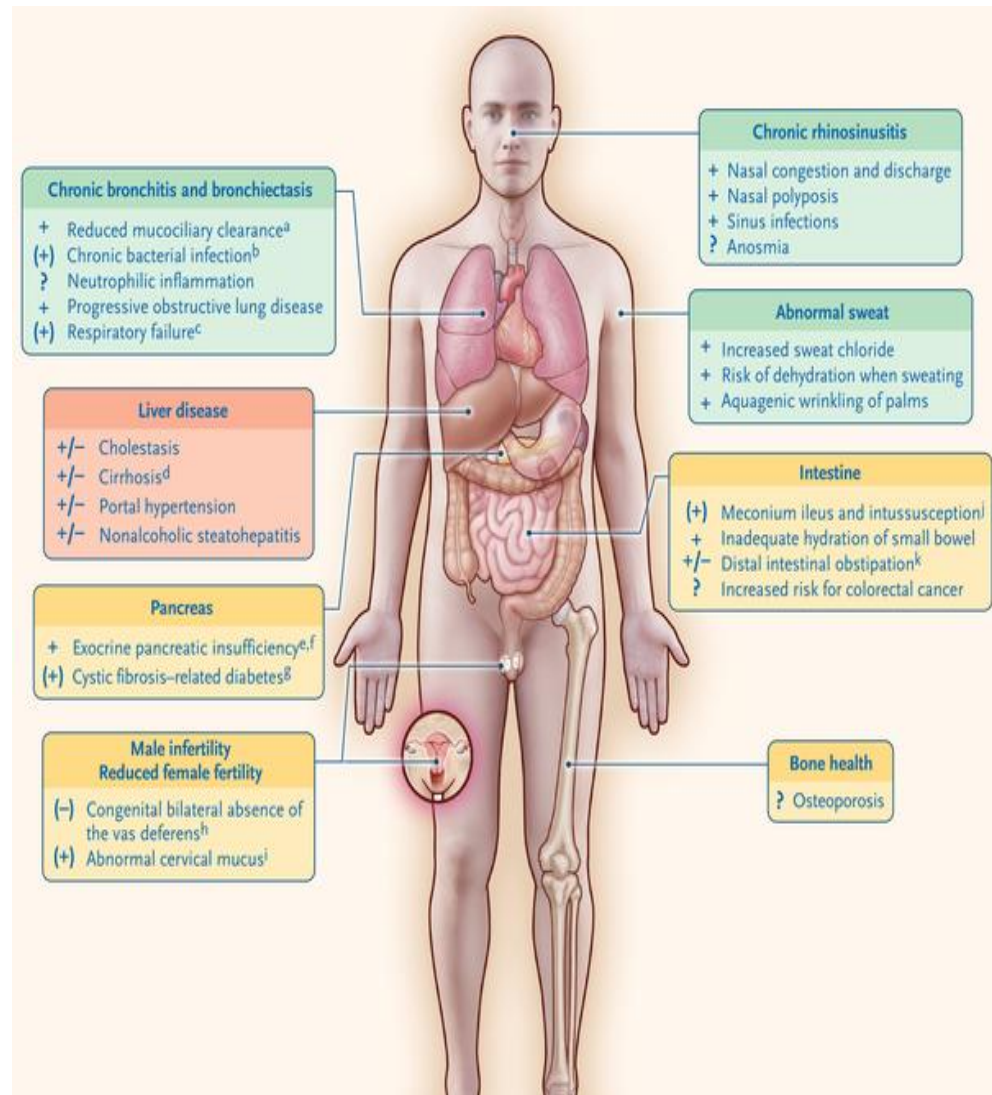
- SCN5A, SCN10A, SLMAP and TRPM4*
 - Potassium current (I_K): *ABCC9, KCND2, KCND3, KCNE1L, KCNE3, KCNE5, KCNH2, KCNJ8, SEMA3A*
 - Calcium current (I_{Ca}): *CACNA1C, CACNA2D1, CACNB2B*
 - Miscellaneous/Other: *HCN4, HEY2, LRRC10*
- Some Drugs to be avoided** (brugadadrugs.org):
 - Antiarrhythmic drugs*: Ajmaline, Allapinine, Ethacizine, Flecainide, Pilsicainide, Procainamide, Propafenone
 - Psychotropic drugs: Amitriptyline, Clomipramine, Desipramine, Lithium, Loxapine, Nortriptyline, Oxcarbazepine, Trifluoperazine
 - Anesthetics / analgesics*: Bupivacaine, Procaine, Propofol
 - Other substances: Acetylcholine, Alcohol (toxicity), Cannabis, Cocaine, Ergonovine

Slide – 29: Cystic Fibrosis

- **Map displaying the estimated diagnosed CF patient burden around the world** (Journal of Cystic Fibrosis Volume 21, Issue 3, May 2022, Pages 456-462)



Slide – 30: Cystic Fibrosis – Clinical Features (N Engl J Med 2023;389:1693-1707):



Slide – 32: Cystic Fibrosis Carrier Screening

Basic and expanded screening panels for cystic fibrosis

Standard or basic pathogenic variant panel
F508del-CFTR, I507del, G542X, G551D, W1282X, N1303K, R553X, 621+1G>T, R117H, 1717-1G>A, A455E, R560T, R1162X, G85E, R334W, R347P, 711+1G>T, 1898+1G>A, 2184delA, 3849+10kbC>T, 2789+5G>A, 3659delC, 3120+1G>A
Reflex tests
I506V*, I507V*, F508C*, 5T/7T/9T¶

CF: cystic fibrosis.

* Benign variants. This test distinguishes between the F508del-CFTR pathogenic variant and these benign variants. I506V, I507V, and F508C testing is performed only as reflex tests for unexpected homozygosity for F508del-CFTR and/or I507del.

¶ 5T/7T/9T analysis is performed only when R117H is detected for the following reasons: The R117H variant can result in classic CF only if there is a 5T variant on the same chromosome and another CF variant on the opposite chromosome. R117H on 1 chromosome and 5T or 7T on the opposite chromosome or homozygosity for 5T (5T on each chromosome) can result in congenital bilateral absence of the vas deferens.

Data from:

1. American College of Medical Genetics. Technical standards and guidelines for CFTR mutation testing. 2006.
2. Committee on Genetics, American College of Obstetricians and Gynecologists. ACOG Committee Opinion. Number 325, December 2005. Update on carrier screening for cystic fibrosis. Obstet Gynecol 2005; 106:1465.
3. Castellani C, Cuppens H, Macek M Jr, et al. Consensus on the use and interpretation of cystic fibrosis mutation analysis in clinical practice. J Cyst Fibros 2008; 7:179.

Graphic 65637 Version 9.0

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Slide – 33: CF Treatment**Medications**

- **Mucolytics**: Dornase alfa/DNAse, Hypertonic Saline
- **Bronchodilators**: Albuterol
- **Antibiotics**: Oral, IV, Inhalation
- **CFTR Potentiators**: Ivacaftor (Kalydeco) for CFTR mutations where CFTR protein is present but Cl⁻ gate “locked” → Opens locked channel (38 FDA-approved mutation genotypes)
- **CFTR Modulators**: Lumacaftor (Orkambi) for CFTR mutations where protein stuck in Endoplasmic Reticulum (ER) → Increases transport of protein out of the ER (F508del Homozygotes)

Slide – 36: AAT DEFICIENCY DIAGNOSIS**Slide – 37: AAT DEFICIENCY MANAGEMENT:**

- **Graph of cumulative survival from Larsson C. Acta Med. Scand. 204:345-351, 1978**
(<https://www.priory.com/cmof/causesof.htm>):
 - A well-established risk factor for COPD is deficiency of the protective protease inhibitor, Alpha-1 Antitrypsin (AAT). This is an inherited autosomal recessive (designated PiZZ) disorder which is fairly rare in the general gene pool. The estimated prevalence of the heterozygote phenotype (PiMZ) is about 2-3% of the Western population. The incidence of homozygous births is in the region of 1 in 3000 live births. As such, AAT deficiency accounts for probably less than 5% of all cases of COPD.
 - Low levels of AAT allow the uninhibited action of elastase on the lung parenchyma, giving rise to destruction of the alveoli and the eventual development of emphysema rather than chronic bronchitis. The pattern of emphysema in AAT deficiency differs slightly from that of smoking-induced pure emphysema in that AAT deficiency produces panlobular emphysema affecting the lower lung fields predominantly, and smoking-induced emphysema is usually centrilobular, affecting the upper lung fields initially. Not all people with PiZZ have very low levels of AAT. Only serum levels below 35% normal are at risk of developing emphysema. Subjects who are PiMZ may also have reduced levels but not as low as PiZZ, and are not thought to be at more risk from developing emphysema than PiMM subjects. Subjects who are PiZZ may not progress to have emphysema, but PiZZ subjects who smoke have a greatly increased risk of developing

emphysema, especially at an early age (Fig 5).

